Theory Note 2 of 8

What to Look For in DNS, HTTP, HTTPS/TLS, ARP, ICMP and SMB

Network Traffic Analysis, Nmap and Windows Logging | Week 2

1. Overview

Every protocol has a typical rhythm: who talks, how often, how much, and in what format. An analyst does not need to memorise every field. What matters is recognising when a protocol behaves differently from its usual pattern. The table below sets out what normal looks like for six protocols that appear in almost every investigation, and what should raise suspicion.

2. Protocol behaviour at a glance

Protocol	Normal behaviour	Suspicious behaviour
DNS (port 53)	Short query and response pairs for known domains. Queries go to the company resolver. Most replies are NOERROR. Volume per host is modest.	Very long or random-looking subdomains (tunnelling or domain-generation algorithms). Many NXDOMAIN replies. Large TXT responses. Queries sent directly to outside resolvers. One host making thousands of requests. Rare or newly registered domains. Zone transfer (AXFR) attempts over TCP 53.
HTTP (port 80)	GET and POST requests to known sites. Mostly 200, 301 and 304 responses. Common browser user agents and ordinary page sizes.	Requests to a raw IP address. Tool user agents (curl, python-requests, sqlmap, Nmap). Repeated 401, 403 or 404 responses (brute force or directory scanning). Credentials in clear text. Large encoded POST bodies. SQL injection or script strings in URLs. Requests repeating at fixed intervals (beaconing).
HTTPS / TLS (port 443)	Valid certificates from trusted authorities. Modern TLS 1.2 or 1.3. Server name (SNI) matches the certificate. Traffic goes to well-known services.	Self-signed or expired certificates. Missing SNI or a mismatch with the certificate. Outdated protocol versions. Unusual JA3 fingerprints. Long sessions with small, regular packets (command-and-control). TLS on non-standard ports. Large uploads to unknown hosts.
ARP	A who-has request followed by a single is-at reply. Each IP address maps to one MAC address. Mostly seen when a host joins the network.	Two MAC addresses claiming one IP. Floods of gratuitous ARP. Replies nobody asked for. The gateway IP suddenly tied to a new MAC (ARP spoofing, man-in-the-middle). One host sweeping the whole subnet.
ICMP	Echo request and reply for troubleshooting. Destination-unreachable and time-exceeded messages from traceroute. Small payloads of 32 to 64 bytes.	Ping sweeps across a subnet. Oversized echo packets or payloads with data (ICMP tunnelling or exfiltration). ICMP redirects. Flood-level volume. Echo requests to many external hosts.
SMB (port 445)	Authenticated file-share access inside the LAN, mostly from workstations to file servers, at a steady and predictable volume.	Workstation-to-workstation SMB. Access to ADMIN$, C$ or IPC$ shares. Many failed logons (password spraying). SMBv1 in use. Large copies at odd hours. PsExec-style service creation. SMB crossing the internet boundary, which should be blocked.

3. Practical tips

- Judge traffic against the baseline for that particular network. What is normal at one site may be an alert at another.
- Look beyond single packets. Volume, timing, direction (internal to external) and rarity usually tell the story.

- In Wireshark, Statistics > Protocol Hierarchy, Conversations and Endpoints give a quick overview, and Follow TCP Stream shows the content of a session.
- Always combine the evidence: a strange DNS query followed by a new TLS connection to the same IP is far more meaningful than either on its own.

> **Key takeaway**
> Suspicion comes from deviation, not from any one protocol being bad. Know the normal rhythm of DNS, HTTP, TLS, ARP, ICMP and SMB, and the odd ones stand out.